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ARTICLE

Classroom Observations and Post-Observation Conversations: Powerful Tools for Developing Teacher Adaptive Expertise

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Abstract
Adaptive expertise is considered key to improving teacher practice. While literature suggests that classroom observations followed by professional learning conversations may contribute to the development of adaptive expertise, few studies have specifically examined the influence of this process. Hence, this qualitative study investigated the perceptions of four teachers and four middle leaders from two New Zealand primary schools, on the use and usefulness of existing classroom observation and post-observation conversation processes in contributing to teacher adaptive expertise. Data included video-recorded observations, pre-observation sheets, transcripts of post-observation conversations and interviews with middle leaders and teachers. Thematic analysis showed that participants believed the process was useful in aiding the development of some elements associated with adaptive expertise such as reflective practice, engagement with evidence, and a focus on improvement. However, data also indicated that a stronger emphasis on goal setting, more time to embed the process in whole school practice would improve its usefulness.

Keywords Adaptive expertise · Reflective practice · School culture · Classroom observations · Professional learning conversations · Post-observation conversation

Introduction

Given that teaching is both complex and demanding, it is important to understand the support teachers require for continuous improvement. Developing teacher adaptive expertise (AE) is regarded as key to improving teacher practice. When teachers develop AE, they engage in reflective practice (RP) and develop critical knowledge and understanding that support them to adapt and change practice to positively

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influence student outcomes (Beltramo, 2017). AE is a key feature of teacher professional learning in New Zealand (Ministry of Education, 2014) with many New Zealand researchers being recognised as experts in this field (e.g., Earl & Timperley, 2008; Timperley & Twyford, 2022). Evidence suggests that classroom observations (CO) and professional learning conversations (PLC) can influence improved teacher practice (Lawson, 2011) and be a catalyst for developing AE (Le Fevre et al., 2020). However, few studies have specifically examined the influence of CO and PLC on developing adaptive expertise. Hence, this research aims to address this gap in the context of New Zealand primary schools, by examining teacher and leader perceptions of the use and usefulness of existing practices of CO and PLC in contributing to AE.

Teacher Adaptive Expertise

Adaptive expertise (AE) is considered a key attribute of quality teaching (Sawyer, 2006). Teachers who have AE are able to find solutions to complex educational problems (Soslau, 2012). An early definition of AE put forward by Hatano and Inagaki (1984) derived from ideas by Piaget (1950) argues that observing the effects of repeated modification of one's skill being applied to varied environments, leads to the development of AE. Liston and Zeichner (1991) define AE as the ability to reflect and refine one's teaching practices, learn from personal experiences to transform future practice. Soslau (2012) posits that AE is evident in teachers who possess effective pedagogical skills, engage in self-assessment of their practice, and adjust their teaching decisions accordingly. Furthermore, AE can be defined as a process whereby teachers use their in-depth knowledge and skills to respond flexibly and responsively to the complexities of teaching by using evidence effectively and evaluating the impact of this practice (Le Fevre et al., 2020). AE is thought to influence teachers to "innovate, when necessary, rethink key ideas, practices, and values in order to respond to non-routine inputs" (Lampert, 2020, p. 10). Teachers who exhibit AE display a deep knowledge of content and the strategies required to teach this content (Timperley, 2011). When confronted with challenges, adaptive experts apply their professional knowledge to find alternative strategies that work. They engage in cycles of evidence-informed inquiry and knowledge-building to identify when routine practices no longer work for their students, adapt practice accordingly, and understand how and where to access support (Le Fevre et al., 2020). AE involves teachers constantly reviewing practice, evaluating its effect on students and taking ownership for improving their knowledge and skills (Timperley, 2015).

AE differs from routine expertise which involves learning a core set of skills to improve efficiency and assumes that through support and practice, teachers can become experts (Soslau, 2012). In contrast, the skills to critically reflect and transform teaching practice and describe and explore key aspects of teaching are not necessarily acquired through repeated practice (Farrell, 2014). To achieve AE, teachers are encouraged to work collaboratively while taking ownership of decisions and actions to solve complex problems (Timperley, 2023). For the purpose of this study, the following attributes that contribute to the development of AE will be explored:

Use of evidence to support PLC, teacher engagement in RP, and creating solutions for improvement.

Using Evidence

A key principle of AE involves using evidence to understand the impact of one's practice. This entails close monitoring to see if changed practice is having the desired impact. Schildkamp et al. (2016) explained that teachers should cultivate a disposition of utilising student achievement as evidence of the impact of their teaching. A New Zealand study by Lai and McNaughton (2008) examined teachers' use of literacy achievement data in 48 primary schools. They found that teachers, who collected, analysed, and discussed student achievement data, were best able to isolate practices that needed improvement. A barrier to engaging in such inquiry is teachers' lack of expertise in interpreting and utilising evidence (Lai & McNaughton, 2008). The process of interpreting evidence is complex, requiring teachers to transform evidence into "actionable information" to adjust future teaching (Timperley, 2015, p. 24).

Reflective Practice

Dewey (1933) defined Reflective Practice (RP) as "Active, persistent, and careful consideration of any belief or supposed form of knowledge in light of the grounds that support it, and the further conclusions to which it tends" (p. 6). Others describe it as deliberate, purposeful metacognitive thinking, influencing teachers to take action to improve their skills (Soslau, 2012). While Ghaye (2011) asserted that RP involves deep engagement with one's assumptions, beliefs, and values, Danielson (2015) believes RP requires teachers to effectively interpret evidence to transform teaching and learning. Vaughn and Parsons (2013) added that RP should be an ongoing endeavour by teachers, even when not required to do so. To be effective, RP should focus on recent practice and result in changed practice (Benade, 2015).

Creating Solutions for Improvement

Another key tenet of AE is teachers' commitment to creating solutions for their problems of practice with a view to continuous improvement (Timperley et al., 2020). This process involves acquiring new knowledge as solution focused action plans are developed. For this to occur, schools need to create the conditions necessary for the above practices to be embedded into school routines. Often, solutions for improvement can be neglected by teachers if not prioritised (Le Fevre et al., 2020), therefore, schools should create a culture supporting teacher accountability for creating solutions and the implementing them (Timperley, 2015).

While there appears to be agreement that AE is necessary to improve teacher practice, there does not seem to be consensus on how to develop AE. Timperley (2015) argues that teachers cannot be expected to analyse and adapt practice on their own or with peers who may have limited professional knowledge. However, PLC

could be beneficial in promoting AE when undertaken between a teacher and a more experienced, knowledgeable individual, such as a middle leader (ML) (O'Leary & Brooks, 2014). Thus, the current study focussed on Classroom Observations and Professional Learning Conversations between middle leaders and teachers.

Professional Learning Conversations and Feedback

Professional Learning Conversations are a formal dialogue between teachers and mentors or leaders, addressing aspects of teaching and learning that are considered pivotal to promoting AE (Timperley, 2015). There has been a trend, both nationally and internationally, for teachers to engage in classroom observations (CO) followed by formal conversations with colleagues to improve practice (Lawson, 2011). In New Zealand, CO and post-observation conversations (POC) are common practice used in schools to further the professional learning and development of teachers (Robinson, et al., 2014). However, previous research involving CO and PLC between colleagues did not always result in improved teaching (Timperley, 2015). One reason for the lack of improvement may lie in observers providing general feedback rather than guiding teachers to reflect deeply. For example, a New Zealand study by Robinson et al. (2014) showed that observers providing general feedback while not disclosing concerns to avoid offending, did not lead to RF, collaborative problem-solving, nor changed practice. Soslau (2012) acknowledged in their US-based study between supervisors and student teachers, that the type of discourse and supervision styles between university supervisors and student teachers could facilitate the development of AE if dialogue encouraged critical reflection and justification of teaching decisions rather than telling, recounting or guiding conversations.

Extensive research has examined leaders or peers requesting teachers to change practice deemed problematic during feedback sessions, with little or no input from the teacher. However, very little research has focused on PLC that encourage teachers to reflect, self-identify problems, and co-create solutions; all of which is necessary for AE (O'Leary & Brooks, 2014). Furthermore, many of these studies involved early career teachers rather than experienced teachers. For instance, Timperley's (2010) qualitative study looked at improving PLC after CO and found that some mentors failed to develop action plans with their student teachers, although much advice was given. Mentors also failed to ascertain if student teachers believed the advice helped them overcome their difficulties. Moreover, feedback lacked critical analysis of specific teacher actions. As O'Leary and Brooks (2014) argue, for conversations to be effective, mentors should ensure a collaborative, evidence-informed inquiry stance where the problem is identified, next steps are co-constructed, and changes to practice are enacted. This is possible when relationships of trust and respect exist between the parties.

According to Timperley (2023), effective conversations that encourage AE are dependent on relationships of mutual trust, respect, and just the right amount of challenge. Without the above, conversations could lead to compliance or a lack of change or improvement. Bryk (2002), determined that in schools with high levels of relational trust, teachers and leaders were more likely to experiment with new

practices while working collaboratively with all stakeholders to advance improvements, all necessary ingredients for the development of AE.

Furthermore, effective PLC have the potential to improve teacher practice especially in conversations where teachers integrate evidence of student learning, evidence of their practice, and other related sources. These sources of evidence provide them with reasoning and guidance for next steps, crucial for the development of AE (Le Fevre et al., 2020). A key step towards AE is teachers moving from passive enactors of improvement to more active agents of change, taking agency for professional development (Timperley, 2015).

Purpose of this Study

While several studies focused on feedback, few have examined how PLC contribute to the development of AE. Furthermore, the studies reviewed in PLC practices involved student or beginning teachers, rather than experienced teachers. Thus, little is known about how CO and PLC could aid the development of AE in more experienced teachers. Therefore, this research involved teachers with a minimum of 5 years of experience. Moreover, most studies on PLC have explored the quality of leader feedback rather than teacher involvement in the reflective process and co-construction of next steps. Finally, much of the research on PLC involved peers and colleagues observing each other, while this study involved ML observing experienced teachers in conjunction with PLC in developing AE. For the purpose of this study, PLC will be referred to as post-observation conversations (POC) a term used by Timperley (2011) to refer to PLC following CO. Hence, the question guiding this study was: What are ML and teachers' perceptions on the use and usefulness of CO and POC to encourage AE?

Method

This interpretive, qualitative study aimed to gain insight into the complex descriptions and interpretations of CO followed by POC and their effect on the development of teacher AE, through the voices of participants. The study was conducted in schools as the natural setting where such POC occur, and drew on interview, audio- and vide-recordings to enable the "discovery, insight, and understanding from the perspectives of those being studied" (Merriam, 2009, p. 1).

Participants

The participants involved four ML and four classroom teachers focusing on literacy, from two large primary schools in Auckland (see Table 1 for participant demographics). The first criterion for identifying leaders was them having responsibility for a team of at least three teachers. Such leaders are usually referred to as ML in the New Zealand context. The second criterion was leaders

Table 1 Participant demographics

School	Middle leader (pseudonyms)	Teacher (pseudonyms)	Teaching experience in years (as ML)	Ethnicity	Gender	Year level
A	Annie		16 (ML 9)	Pasifika	Female	3–4
	Adele		15 (ML 4)	Indian	Female	5–6
		Cassie	5	Middle Eastern	Female	3–4
		Chloe	5	NZ European	Female	5–6
B	Belle		11 (ML 5)	NZ European	Female	1–2
	Brian		14 (ML 2)	NZ European	Male	4
		Drew	7	Māori	Female	2
		Derek	19	NZ European	Male	4

being experienced in conducting CO and POC. Thirdly, because this research focused on experienced teachers, participants had to have at least 5 years of teaching experience. Finally, all teachers involved had to teach literacy. Literacy was chosen because it is considered a core curriculum area in primary schools.

Measures

This study included four measures: (1) pre-observation sheets, (2) video-recorded CO, (3) POC audio-recordings, and (4) semi-structured individual interviews with the ML and teachers.

First, a pre-observation sheet developed by the researcher was completed by teachers in a meeting with their ML prior to the CO. Teachers recorded their teaching goal as well as associated success criteria for the lesson. Pre-observation sheets were utilised by the researcher to compare teachers' goals and relevant practice when analysing CO videos. Second, ML observed and video-recorded a 20-min lesson. The recordings of the CO enabled the researcher to compare what participants had intended to improve in their teaching with their actual classroom teaching. Furthermore, it supported the researcher's understanding of the POC. Third, each ML audio-recorded the POC which ranged from approximately 5–20 min in duration. The recordings were transcribed by the researcher and provided a core data source to examine how ML supported teachers to engage in practices that supported the development of AE. Finally, 1-h semi-structured interviews were held with each teacher and ML after the POC to gather their perspectives on the process. Semi-structured interviews allowed for participants when probed, to provide rich insights into their perspectives. The interview guide was piloted to ensure it was well constructed and fit for purpose with a teacher who did not participate in this study (Gubrium et al., 2012).

Data Analysis

Data was analysed using thematic analysis and the six stages proposed by Braun and Clarke (2006). Firstly, the researcher became familiar with the data through repeated reading and review, noting potential patterns and meanings. Each recorded CO, interview and POC transcript was reviewed simultaneously to check for accuracy (Braun et al., 2019). Secondly, codes were generated, and data was organised around commonalities and differences in the use and usefulness of CO and POC. Initial themes included the degree teachers engaged in RP, areas identified for improvement, and action taken to improve practice. The researcher adhered to the seven attributes for coding proposed by Saldaña (2009): high degree of organisation; perseverance; the ability to deal with ambiguity; remaining flexible; creativity; maintaining high ethical standards; and using the right words for the codes or themes.

Thirdly, codes were sorted into potential themes associated with features of AE, and coded data extracts were collated within the identified themes. Comparisons were made and patterns identified. Thereafter, themes were reviewed using a thematic map at the case level (each leader–teacher pair), followed by a cross-case analysis, where the data and coding were brought together for the entire data set. Next, some themes were merged due to overlap, some were separated, and others were discarded due to a lack of supporting data. Final themes for developing AE through CO and POC were: school culture, developing RP, engaging with evidence, and creating solutions. Additionally, requiring more time and setting goals were identified as themes to strengthen the process.

The peer debriefing strategy proposed by Creswell (2014) was employed to enhance the accuracy and trustworthiness of the findings. Coding was examined by both co-authors who randomly coded a small sample of the data. Final themes were agreed upon by the authors, after which all coding was reviewed.

Findings

Overall findings show that ML and teachers perceived the process as contributing to improved teaching and learning, and all teachers reported a shift in classroom practice. Conditions leading to the development of AE seemed more prevalent in School B where CO and POC processes were embedded in school culture. There were commonalities and differences in the ML and teacher perspectives on the use and usefulness of CO and POC. Four themes for enablers of AE were identified: school culture, developing RP, engaging with evidence, creating solutions. Additionally, providing more time and setting goals were thought to be vital to improving the CO and POC process. Table 2 provides an overview of the findings by these themes by school. These are then further unpacked and quotes from participants provided.

Table 2 Overview of identified enablers for adaptive expertise by school

Themes	School A	School B
School culture	CO and POC not embedded in the school culture Happened infrequently Teachers were anxious and daunted by the process Feedback focused with the teacher as a passive participant Little to no challenge from ML	CO and POC was embedded practice Frequent (4 times a term) Teachers at ease and perceived it as a positive experience Effective questioning leading to greater teacher reflection and contribution to the conversations Relationships of trust allowed for ML to respectfully challenge teacher actions
Developing reflective practice	Conversations were not a collective endeavour -ML provided feedback with little or no opportunity for teachers to share their thinking	Effective ML questioning led to teachers evaluating their teaching actions and identifying ways to improve—allowing teachers to look deeply into their thinking and decision-making
Engaging with evidence	Did not engage with evidence to support evaluations made by ML—reason might be the lack of a teacher goal and co- constructed success criteria	Evidence was used and interpreted to support conclusions drawn about the quality of teaching use of data brought a focus to the conversation Teacher goal and success criteria co-constructed
Co-constructing solutions	Solutions to problems of practice/next steps were identified and stated by ML Participants believed the process led to improved practice	Solutions to problems of practice/next steps were co-constructed Collaborative exercise Participants believed the process led to incorporating new approaches to teaching
More time	Little or no time allocated to pre-observation conversations Little or no focus on teacher goal CO and POC short and sometimes interrupted	Time allocated for co-constructed teacher goal during pre-observation conversations Participants felt that more time was needed in each step of the process to ensure deep reflection and learning for teachers
Setting goals	Meeting to establish teaching goals and associated success criteria were not always possible as the school seldom prioritised this step in the process Focus on compliance rather than meaningful conversations focused on developing AE	Teacher goal and associated success criteria were co-constructed before the CO These were the focus for the POC and lead to co-creation of next steps for improvement Provided conditions that contribute to the development of AE

School Culture

School culture refers to everyday practices used by all staff consistently. Both schools differed in their use of CO and POC. In School B, CO and POC were part of their ongoing practice, conducted at least four times a term. Both teachers reported being comfortable with the process. Drew (Teacher, Interview) described her experience as "really daunting when I first started, but we get observed every 2–3 weeks… I've become accustomed to it". Similarly, Derek (Teacher, Interview) said, "I hated being observed, now it's part of our school culture, always having someone come and observe us, so I really don't mind it". Teachers seemed to feel supported as indicated by this statement, "We're a team. Everyone is open to having their practice observed and analysed. We do it often, so we're used to it" (Drew, Teacher, interview). Derek articulated a similar observation, "Our staff are on a learning journey, I don't feel threatened. We are observed often and see it as a professional development opportunity. (Interview). This being a supportive process was evident in the CO and POC. Teachers appeared relaxed during teaching, Derek in particular, interacted with his students quite naturally. Furthermore, Drew and Derek were active participants in their POC. In fact, Drew started the POC with, "… since my goal was to involve students more in the discussions, I should have done less of the talking, I should have encouraged them to use talk moves more". She identified an area for improvement before receiving feedback from her leader. ML in School B shared reasons why they believed teachers were open to observations and conversations. Belle (ML, Interview) felt it was the "way the school works… the setup, open plan classrooms and we have so many visitors". Similarly, Brian (ML, Interview) said "It's part of what we do", implying that they were an embedded feature of the school culture.

In contrast, School A conducted CO once a term and teachers seemed anxious about the process. Cassie (Teacher, School A, Interview) noted that CO were "nerve-wracking", appearing nervous during her lesson and contributing very little to the POC. Similarly, Chloe (Teacher, School A, Interview) stated, "I felt really self-conscious. I didn't like someone watching me". When asked how the process might be improved, she responded, "These should be held more often, I can't think of a better way to get accustomed to it and not freak out". Her comment aligns with Drew's (Teacher, School B, Interview) quote above which is that one has to get used to the process to find value in it.

Thus, findings suggest a connection between the embeddedness of the process within the school, teachers' ease, and its perceived value. Furthermore, six of the eight participants felt that relationships between ML and teachers were central to the success of this process. They believed that values of respect and trust helped ML to challenge teachers, promoted teacher honesty about practice and encouraged them to take on board suggestions for improvement which is vital for the development of AE. This was highlighted by Brian (ML, School B, Interview) "There's a lot of respect for one another, no one feels they are being judged. I don't want to break their trust. I approach them in a supportive way". Interestingly, Cassie (Teacher, School A, Interview), also said her relationship with her ML helped her be open and honest, "My team leader is easy to work with, so understanding. She respects

my opinions, puts me at ease, and I'm comfortable being open and honest with her." The above examples indicate that CO and POC were more successful in the school where these practices were embedded as opposed to the school where it was not embedded practice.

School culture seemed to influence how ML conducted their POC to encourage AE. In School B where CO and POC was embedded, there was greater evidence of RP, engagement with evidence to support conversations, and collaborative co-construction of solutions. Furthermore, time was set aside for ML and teachers to meet for a pre-observation conversation to discuss and set a teacher goal for the CO while in School A, meeting prior to the CO was not a usual occurrence.

Developing Reflective Practice

It was revealed that CO followed by POC encouraged teachers to become more reflective. ML typically described teachers' ability to reflect with statements such as "It's helped even the least reflective teacher become more reflective by looking deeply into what worked and what didn't" (Adele, ML, School A, Interview), while Brian (ML, School B, Interview) noted that "…[it] forces teachers to take notice of their actions and impact on their kids' learning, to think about the effect of your teaching on your class…". Belle (ML, School B, Interview) believed that CO and POC encouraged teachers to evaluate their practice and identify ways to improve, stating "I don't know a way more conducive to teachers problem-solving than discussing it".

ML felt that questioning teacher actions and the reasoning behind them, was instrumental in growing their RP by encouraging them to review their actions. However, examples of such questions were only evident in the POC in School B, "Why did you choose this strategy to engage Mary?" (Brian, POC), and "How did you think the lesson went?" (Belle, POC). Both ML from School A provided little or no opportunity for teachers to share their thoughts during the POC. For example, Annie (ML) told Cassie what improvements she should make to her future teaching, giving Cassie little opportunity to express her own views on the lesson. This POC consisted predominantly of Annie's feedback. Similarly, Adele (ML) provided feedback for seven minutes before allowing Chloe to share her thoughts briefly. The POC provided little evidence that the process supported RP.

All participants in School B perceived CO and POC with experienced leaders to be more valuable than individual reflection. Brian affirmed that while self-reflection was beneficial, there was greater value in reflecting through discussion with an experienced practitioner, saying, "Individual reflection works to a certain extent but discussing with someone very knowledgeable gives greater clarity" (Brian, ML, Interview). Brian believed experienced observers paired with teachers, improved RP. This was echoed by Derek (Teacher, Interview), "It's a good way to reflect… assess how you are going with some expert help".

Despite two of the four POC showing little evidence of RP, all four teachers believed they had grown as reflective practitioners. Cassie from School A said the process encouraged RP and improved teaching, noting, "I have a chance to look back

on my teaching, what needs improving, then change things accordingly". Similarly, Chloe acknowledged, "It prompted me to evaluate what I currently do and think of ways to improve" (Teachers, Interview).

Engaging with Evidence

It proved a challenge for some participants to engage with evidence when identifying and focusing on practices for improvement. Statements such as "I think the lesson went well" (Drew, Teacher, School B, POC) or "I'm okay with how the lesson turned out" (Chloe, Teacher, School A, POC) were not supported with evidence for conclusions drawn, nor did their ML ask for evidence to support their thinking. Despite Chloe and Adele (ML, School A) completing a pre-observation sheet outlining success criteria, these were not discussed during the POC. While Drew did not use evidence to draw conclusions about her teaching, her ML however, was able to refer to specific evidence regarding the impact of her teaching as indicated below:

> "Your goal was to give students more opportunities to participate in the discussions. When you asked students for describing…you immediately called Ben
> to answer. He didn't have a response despite your prompting. Maybe allow students
> *Some think-time before asking, or even pair-share."* (Belle, ML, School B, POC).

Belle's actions seemed deliberate, saying in the interview, "I try to draw conclusions based on the evidence from the lesson and impact of teaching on student learning". Also from School B, Brian (ML) and Derek (Teacher) engaged in a POC informed by a range of evidence. Brian began by re-stating Derek's goal which was to scaffold reluctant writers to write complex sentences. Brian asked, "How do you know the sentence starter strategy helped John?" Derek responded, "He wrote far more…the sentence beginnings relieved some of that pressure". However, Derek also admitted that while he wrote more than usual, John had not achieved the goal of creating a complex sentence. Thus, both pairs from School B were able to draw on student participation as evidence of teacher effectiveness in contrast to School A where ML comments remained general in nature.

Co-constructing Solutions

Analysis of POC indicated an improvement focus for all POC, whether it was feedback directed to the teacher or co-constructed by the pair. In all instances, conversations involved creating solutions to problems of practice. All participants noted in the interviews that the process encouraged knowledge building and changed practice. Teachers believed that they learned new approaches to teaching and extended their repertoire of teaching strategies. Drew (Teacher, School B, Interview) felt, "It helps evolve practice… include new and different skills that have a positive impact." Her ML Belle (Interview) concurred, "We always introduce a new teaching strategy

or approach for the teacher to trial". Similarly, Chloe (Teacher, School A, Interview) mentioned, "I learn a lot about my teaching, we look at new skills that I can develop". Her ML Adele confirmed that her teachers "appreciated being supported with new approaches" (Interview). Teachers commented in similar ways: "I've learnt many new ways to address my students' learning needs" (Derek, School B, Interview), and "Each time I learn something new that builds my skill set." (Chloe, School A, Interview). All four teachers believed that CO and POC helped them grow and develop their practice.

Despite all ML suggesting that creating solutions to teachers' problems of practice was a collaborative process, POC showed this was not always the case. Both ML Annie and Adele from School A, provided the solutions rather than co-constructed them with their teachers. Adele explained to her teacher "…I think they [students] were a bit confused. You need to model the task before getting them to complete it." (School A, POC) All four teachers said they had implemented the new skill or strategy agreed upon during the POC. In Drew's case, a link between her next steps discussed and implementation of the new teaching approach was evident: "… we worked together to identify my next steps, we worked out a brief action plan. I've implemented this and gauged its impact" (Teacher, School B, Interview) Similarly, Chloe explained, "We look at next steps…I implement the new skill …and reflect on it" (Teacher, Interview). Chloe was the only teacher to mention a follow-up meeting with her leader to discuss the impact of this new practice.

More Time

Participants from both schools preferred more time for the process. In School A, pre-observation conversations were not a usual occurrence. ML revealed that time constraints sometimes did not allow them to meet before a CO. Annie (ML, Interview), noted, "We had no time to meet before the observation". Adele (ML, Interview) admitted that they "…had a very brief discussion for lack of time", and she was, "Not sure this was enough, but we discussed what she wanted me to observe". It appeared that there was little importance placed on pre-observation conversations and therefore very little or no time allocated to meet. Teachers believed that more time for pre-observation conversations was beneficial, "We had a brief meeting, I'd have preferred a specific goal for my observation". (Chloe, Teacher, Interview). Cassie (Teacher, Interview) remarked, "We didn't meet beforehand, I preferred a writing observation to get feedback on modelling writing". Her ML had observed a reading lesson instead. Typically, the responses, especially from participants in School A, identified time constraints as a barrier to engaging in effective pre-observation conversations.

Three ML felt that more time dedicated to POC meant that more questions could be posed to surface teachers' views and allow for authentic collaboration. Belle (School B, Interview) stated, "More time will allow for more questions to understand the thinking behind teachers' decision making". Annie (ML, School A, Interview) said that more time would, "make it a collaborative exercise…but because of lack of time, I provide the feedback and action plan."

Similarly, Drew (Teacher, School B, Interview) felt that more time for the POC would help her better understand how to implement her new goal. She noted, "We're always in a rush to finish within the set time frame…it's not always clear how I go about achieving my new goal". Overall, one pair from School B and all from School A identified lack of time as a challenge to engaging in meaningful POC.

Setting Goals

Identifying a teaching goal for the CO was considered to be important by seven of the eight participants. Chloe thought it was important to discuss her goal so she knew what her team leader would be focusing on rather than, "spending too much energy wondering what she was going to be looking for". (Teacher, School A, Interview), and others reiterated this sentiment, "Having a goal is important, you want them to see if you're actually achieving it. Otherwise, there's no point" (Cassie, Teacher, School A, Interview). Drew (Teacher, School B) said, "…if I didn't know what she was coming in to observe, I wouldn't be able to show case that skill". Three of the four ML felt that goal setting was important as a focus for the CO, ML Brian noted:

"It's very helpful to ask the teacher what they would like me to focus on during the observation otherwise it becomes a guessing game… her wondering what I'm going to be looking for and me hoping that I focus on aspects she would like to improve." (School B, Interview).

Meanwhile, Annie (ML, School A) believed that CO did not have to focus on a singular aspect of a teacher's practice to initiate a meaningful POC. She felt "…it doesn't matter, you are always going to focus on some aspect of teaching, and that's what you discuss afterwards". Her teacher Belle, however, found that when she had "gone into a CO without agreeing on a goal it made it harder to have a focussed discussion" (Interview).

An analysis of POC indicates that they all had an improvement focus, whether in the form of leader feedback or a two-way conversation between ML and teacher. A clear connection between the teacher goal, key teaching strategies observed in the lesson and discussed in the POC, could be identified in School B. For example, from Drew's CO video it was evident that her goal was to improve her scaffolding strategies. During the POC, they discussed the impact of these strategies on student learning and addressed areas for improvement. In contrast, the obvious teaching focus in the lesson did not feature in the POC between Annie and Cassie from School A. Participants agreed that co-constructing teaching goals before the CO improved POC.

Discussion

This study examined the use and usefulness of CO and POC in the development of AE, as perceived by ML and teachers. Findings suggest that CO and POC were seen as a catalyst for the development of AE. Where CO and POC was embedded in school culture, incremental shifts in line with the development of AE was evident.

The process seemed to encourage teacher RP, influence engagement with evidence, and engender the co-creating solutions for improvement. Furthermore, prioritising time for the process supported teachers to co-construct goals and success criteria for the CO and engage in meaningful POC, thus contributing to AE.

Embedded Practice

Both schools used CO and POC as a means to promote teacher professional learning. However, they each differed in how this process was used to achieve its aim. In School B, CO and POC seemed to be embedded practice, occurring four times a term. There seemed to be a correlation between the frequency of CO and POC and teachers' ease with the process, perceiving it as a positive experience. Frequent observations and conversations made teachers less anxious and more comfortable with the process, thereby reducing uncertainty and vulnerability, allowing them to express themselves and trial new ideas, necessary ingredients for growing AE. Teachers described the process as useful in supporting their learning, however, literature confirms that CO and POC do not happen by themselves but require school leadership to prioritise time for frequent professional learning or its impact may be limited (Robinson et al., 2014).

In School A where CO and POC occurred once a term, teachers were anxious, stressed and daunted by the process. They suggested these be held more frequently so they could become accustomed to the process. Timperley et al. (2020) posit that the frequency of meeting to engage in professional learning conveys tacit messages to staff about what is valued, which in turn, influences how teachers react. There is correlation in a New Zealand study in the primary school context between the frequency of CO and teaching quality, with greater shifts in teacher practice occurring when they were observed more frequently than schools where teachers were observed occasionally (O'Leary & Brooks, 2014).

Another characteristic of embedded practice was relationships of trust and respect between ML and teachers. CO and POC were approached with ease in School B as opposed to School A where they seemed to be tolerated. All four teachers believed that trust and respect were necessary for them to be honest about their problems of practice. Well established relationships allowed for respectful challenge of teacher practice, an important ingredient for the development of AE as explained by Timperley (2023). Teachers and leaders who relate well to one another share knowledge, expertise, and participate in peer critiquing thus positively contributing to improved teacher practice, but the status quo remains if practices are not respectfully challenged (Timperley, 2011). The importance of relationships was highlighted by all participants in this study, yet the level of challenge presented in POC can be described as low for School A. The POC were of a shorter duration, ML provided feedback with little opportunity for teachers to co-create next steps collaboratively, and initial conversations for goal setting were not prioritised. Where practices were embedded within the school and relationships established, teachers were more open to suggestions for improvement and more likely to embrace them, creating the conditions that lead to AE.

Developing Reflective Practice

Reflecting on practice within a learning community is considered a necessary practice for building AE (Timperley, 2013). Findings from this study indicated that CO and POC seemed to have a positive effect on teacher RP, especially when the process was embedded in school culture. Participants believed it encouraged them to recognise recent actions and its impact on student learning. Teachers not only evaluated their practice, but also identified ways to improve. This aligns with Benade (2015), who posited that RP focusing on recent practice and resulting in changed practice, is beneficial for teacher growth and development, making CO and POC an effective vehicle for the development of AE.

However, the key to encouraging teacher RP seems to lie with ML encouraging teachers to inquire through questioning rather than merely providing feedback. When asked for reasons behind their decision-making, teachers were able to reflect on teaching actions and propose improvements. Ghaye (2011) found that questioning teachers encouraged them to reflect on their actions which usually resulted in improved outcomes.

This study highlighted the importance of RP being collaborative rather than an individual pursuit and preferably, with experienced, knowledgeable practitioners. Given the complexity of reflecting and adapting teaching, Timperley et al. (2020) agreed that teachers cannot be expected to do this on their own or with peers having limited professional knowledge. Collin et al. (2013) similarly argue that RP should be a collective endeavour, while O'Leary and Brooks (2014) advocate for experienced practitioners to encourage RP by probing teachers through questioning, to look deeply into their thinking and decision-making.

Engaging with Evidence

Another element essential for AE is using evidence to encourage evaluative thinking during conversations (Timperley, 2023). Evidence refers to any observable data that might support conclusions drawn. While participants in School B engaged with evidence to support conclusions drawn about teaching and learning, School A did not. A reason for this challenge might be the lack of a goal and co-constructed success criteria against which both teacher and ML could gather evidence.

As reported by Schildkamp and Poortman (2015) in a Dutch study, the use of data brings a focus to conversations and promotes the "fertile exchange" of thoughts (p. 2), as was apparent in School B where evidence was used to support conclusions drawn and in co-constructing solutions to the identified problem, key characteristics of EA. Interpreting evidence is vital to problem-solving and improving teaching (Soslau, 2012). Greater collaboration between the pairs encouraged the interpretation of data (Timperley, 2015), however, interpreting evidence is a skill that teachers need support to develop (Timperley & Twyford, 2022). It may be that ML in School A were not aware that this was part of their role, or they may not be adept at interpreting data themselves.

Creating Solutions

POC between ML and teachers were shown to have an improvement focus, whether in the form of feedback given by the ML or a two-way conversation between the pair. Conversations focused on creating solutions and next steps for improvement. Both teachers and ML believed that CO and POC resulted in teachers developing new approaches to teaching. This is in keeping with literature affirming that when new knowledge is brought explicitly to conversations, changes in teaching practice are likely to occur and AE is encouraged (Beltramo, 2017; Timperley, 2011).

Despite Danielson (2015) stressing the importance of collaboration in identifying next steps, all four POC had focused on creating solutions to identified problems of practice, even when it was not a collaborative exercise. However, where CO and POC were embedded in schools, teachers could see how the combined effect of CO and POC cycles resulted in knowledge building and improved practice over time.

Improving Classroom Observations and POC

Setting aside sufficient time for each part in the process seemed a challenge in both schools. Participants believed that more time for conversations before and after CO would allow for deeper discussion and improved outcomes. Finding time for POC is often problematic in schools and ad hoc conversations are usually short and sometimes interrupted (Ghaye, 2011). This was reflected in both schools but more so in School A, where participants spoke of the challenge of finding sufficient time to devote to pre-observation conversations.

Meeting to establish a teaching goal was not always possible due to time constraints, schools not prioritising this step, or it being rushed. Timperley (2023) posits that if there is no clear purpose established before a conversation, confusion may arise or trust can be destroyed, both of these outcomes are at odds with developing AE. In their US-based study, Lotter, et al. (2014) used a set of protocols whereby the coach and teacher discussed the teaching goal and criteria prior to the CO. Teachers' knowledge deepened as a result. Participants in School A who did not set a goal together or co-construct criteria to measure their success, focused their POC more on completing a process rather than forming a platform for meaningful professional learning.

Limitations

This small-scale qualitative study involved two teachers and two leaders from two Auckland schools. Due to limitation of size, the findings provide limited scope for generalisability. A larger study involving rural and urban schools, or small and large schools could provide additional data to support this qualitative study which focused on obtaining rich data to understand the influence of CO and POC in supporting teachers to develop AE. Furthermore, despite teachers reporting a shift in classroom practice, this study was not able to ascertain if an actual shift had occurred. Future

research could include investigating the relationship between the use of CO and POC to improve teaching and student achievement outcomes. Further research could include student views relating to changes in lessons taught following CO and POC.

Conclusion

The aim of this study was to explore the use and usefulness of CO and POC in enabling teachers to develop AE. Although few in number, studies that have explored the role of CO and POC in developing teachers AE, have provided rich insights into how school culture and relationships could embed practices that support the development of teacher AE. The findings from this study emphasise that where school practices are embedded, teacher experiences of CO and POC are positive and ML seem more equipped to support teachers in growing AE through positive relationships and effective questioning. Furthermore, conversations exhibited characteristics that contribute to the development of AE such as improved RP, use of evidence to identify foci for improvement, leading to the creation of new practice, The qualitative data also provided specific aspects that hindered the effectiveness of the process, providing a rationale for why it is crucial that school leadership prioritise time for CO and POC and ensure that it is part of the school culture rather than a solitary exercise. Schools are encouraged to create a culture where individuals become accountable to each other for their ongoing development of AE (Timperley, 2015). It is hoped that this study could be seen as a lead to more in-depth studies aimed at developing AE as a means to improving teacher practice while evaluating its impact on teaching over time.

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Declarations

Conflict of interest The authors have no conflicts of interest to disclose.

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